

BioShoulder: A Large-Scale Biomechanical Dataset for Shoulder Muscle Activations

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Abstract—Real-world human shoulder muscle activation data is critical for advancing upper-limb rehabilitation, sports biomechanics, and human-robot collaboration. However, these data remain scarce because acquiring them is highly resource-intensive and often invasive. Generative simulation methods for muscle activation have helped address this data scarcity by enabling large-scale synthetic data generation. However, their application to shoulder-joint simulation remains limited, as the shoulder’s exceptional degrees of freedom and complex compensatory mechanisms frequently cause severe mathematical divergence and unrealistic artifacts in physics engines. To bridge this gap, we introduce BioShoulder, a large-scale dataset of quality-controlled simulated shoulder muscle activations. Using whole-body motion data and action annotations, we apply trial-level semantic curation, simulation feasibility filtering, and window-level activation screening before extracting bilateral shoulder activations with a musculoskeletal model. BioShoulder contains 1,886 continuous free-space motion trials from 302 subjects, totaling 9.732 hours. These trials are segmented into 48,137 activation clips with 40 bilateral shoulder activation channels. We further establish a sequence-to-sequence benchmark under sequence-level and motion-type splits. Results show that shoulder activation prediction is feasible from motion features, while the performance drop under the motion-type split highlights the challenge of generalizing to unseen primary motion types. BioShoulder provides a public resource for studying motion-dependent shoulder muscle activation and biomechanically informed human motion modeling. Anonymous dataset, official splits, and benchmark code are publicly available¹.

Index Terms—Shoulder biomechanics, musculoskeletal simulation, muscle activation, human motion, biomechanical dataset

I. INTRODUCTION

Understanding how shoulder muscles are recruited during human motion is important for upper-limb rehabilitation, sports biomechanics, and human-robot collaboration [1]. Accurate muscle activation data can reveal the internal biomechanical demands underlying externally observable movements and support the development of muscle-aware human motion models. However, constructing large-scale datasets that pair three-dimensional human motion with shoulder muscle activations remains challenging. Surface electromyography (sEMG) can measure only a limited set of superficial muscles, while intramuscular EMG is invasive and difficult to deploy at scale. Both approaches also require carefully controlled experiments and synchronized motion capture. Consequently, existing experimentally acquired muscle activation datasets generally contain limited subjects, muscles, or movement types [2], [3].

Musculoskeletal simulation provides a scalable alternative for enriching motion capture data with estimated muscle activations. Platforms such as OpenSim [4] estimate the muscle recruitment patterns required to reproduce observed motions under biomechanical constraints. Recent work, particularly Muscles in Time (MinT) [5], has demonstrated that large motion repositories can be augmented with simulated muscle activations and subsequently used to train motion-to-muscle prediction models. Nevertheless, existing large-scale simulated datasets primarily focus on the lower body and trunk, leaving shoulder-specific muscle activation data comparatively under-represented.

Extending this simulation paradigm to the shoulder complex presents additional challenges. The shoulder involves multiple coordinated joints and redundant muscle paths, making inverse kinematics and static optimization sensitive to kinematically infeasible motions and model constraints [6], [7]. Moreover, muscle recruitment during object interaction depends on the external forces and torques transmitted through contact. Although motion capture data may describe the manipulated objects or nominal loads, the corresponding time-varying interaction forces are typically unavailable and cannot be reliably inferred from kinematics alone [8], [9]. Directly simulating contact-intensive motions may therefore produce inaccurate activation estimates, motivating careful motion selection and simulation-level quality control.

To address these limitations, we introduce BioShoulder, a large-scale simulated dataset dedicated to muscles acting on the shoulder joint. Starting from AMASS motion sequences [10], we use action annotations from BABEL [11] to exclude trials involving physical contact, object manipulation, or other substantial external loads. The retained free-space motions are mapped to the MOBL-ARMS upper-extremity musculoskeletal model [6], [12] in OpenSim. Trials that cannot be processed successfully by inverse kinematics and static optimization under the model constraints are discarded, followed by window-level screening for implausible activation saturation. The resulting dataset contains 1,886 continuous trials from 302 subjects, covering 9.732 hours and 30 action categories. After temporal segmentation, it provides 48,137 overlapping motion-activation clips with continuous activations for 40 bilateral shoulder muscle strands, comprising 20 strands on each side.

To assess the utility of BioShoulder for data-driven biomechanical modeling, we establish a shoulder-specific benchmark for sequence-to-sequence motion-to-muscle activation prediction. The benchmark evaluates representative recurrent,

¹<https://anonymous.4open.science/r/bioshoulder-DE32>

convolutional, state-space, and attention-based architectures using predefined data splits and complementary error- and correlation-based metrics.

Our main contributions are summarized as follows:

- **A large-scale shoulder-specific motion-to-muscle dataset.** BioShoulder provides 1,886 continuous free-space motion trials and 48,137 fixed-length clips paired with continuous activations for 40 bilateral muscle strands acting on the shoulder joint.
- **A multi-level data curation pipeline.** We combine trial-level semantic selection, trial-level simulation feasibility checks, and window-level activation screening to reduce errors caused by missing external loads, failed optimization, and activation saturation.
- **Standardized motion-to-muscle benchmarks.** We release predefined evaluation splits and representative sequence-modeling baselines for studying shoulder muscle activation prediction.

II. RELATED WORK

Human Muscle Activity Datasets Experimental muscle activity datasets commonly rely on surface electromyography (sEMG) together with kinematic or visual measurements. Camargo et al. [2] collected multimodal lower-limb biomechanical data across level-ground, ramp, and stair locomotion. For upper-limb tasks, KIMHu [13] provides synchronized kinematic, imaging, and EMG data for human muscular manipulability analysis, but includes only four EMG channels, with only one corresponding directly to a shoulder muscle. Muscles in Action (MIA) [14] synchronizes whole-body exercise videos with sparse sEMG signals from selected upper- and lower-limb muscles, including bilateral biceps brachii and latissimus dorsi but no deltoid or rotator-cuff muscles. MuscleMap [15] further expands the scale of video-based muscle analysis by providing 20 binary muscle-group labels across diverse exercises, but represents the shoulder using only a coarse region-level label. Although these datasets have advanced muscle-aware human motion analysis, they either provide sparse or coarse muscle observations or focus on lower-limb, general upper-limb, or whole-body exercises rather than continuous activation signals spanning the shoulder complex.

Simulation-based Muscle Activation Generation Biomechanical simulation provides a scalable alternative to experimentally acquiring muscle activity across large motion collections. OpenSim-based static optimization estimates muscle recruitment patterns that reproduce motion-derived joint moments under musculoskeletal and optimization constraints [4]. Faldotto et al. [3] combined experimental EMG with physics simulation to analyze lower-limb muscle synergies and their contributions to joint torque. At a substantially larger scale, Muscles in Time (MinT) [5] enriched human motion sequences with simulated muscle activations and established benchmarks for predicting these activations from human pose. However, existing large-scale simulation-based datasets primarily focus

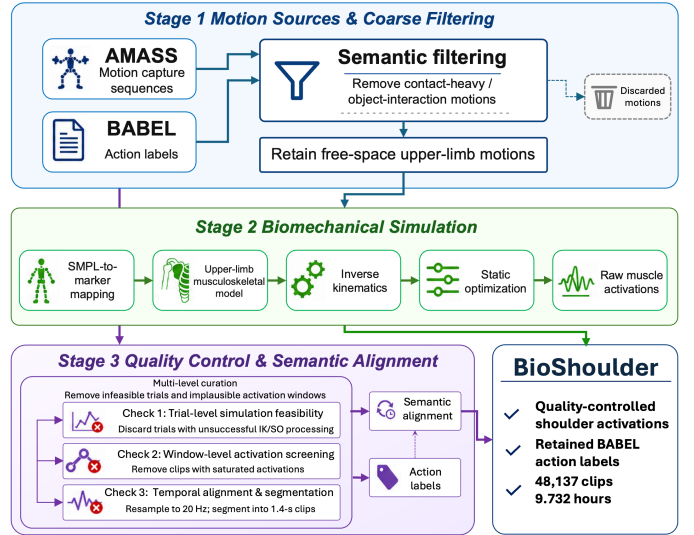


Fig. 1. The overall data generation pipeline of the BioShoulder dataset.

on the lower body and trunk, and do not provide shoulder-specific continuous activations across diverse free-space motions.

III. DATASET CONSTRUCTION

A. Data Source and Biomechanical Simulation

BioShoulder is constructed from AMASS [10], a large-scale motion repository that represents diverse motion capture datasets using the SMPL body model [16]. We use motions from six AMASS sub-datasets: EyesJapan, CMU, BMLrub, KIT, BMLmovi, and TotalCapture. These subsets provide diverse free-space human motions and subject body shapes, forming the kinematic basis of BioShoulder.

To estimate shoulder muscle activations, we map SMPL-based motion sequences to a bilateral MOBL-ARMS upper-extremity musculoskeletal model in OpenSim 4.4 [4], [6]. MOBL-ARMS provides an established anatomical representation of the upper limb and shoulder complex, including the clavicle, scapula, humerus, and associated muscle paths. In our pipeline, SMPL body geometry is converted into virtual anatomical markers compatible with the OpenSim model. The musculoskeletal model is scaled using subject-specific body shape information from AMASS, with body mass estimated from SMPL body volume following Bittner et al. [17].

For each motion trial, we first perform inverse kinematics (IK) to estimate the generalized coordinates that best reproduce the marker trajectories. Static optimization (SO) is then used to estimate muscle activations at each frame by solving for a muscle recruitment pattern that generates the required joint moments under the actuator and joint constraints defined in the MOBL-ARMS model. The SO objective minimizes the sum of squared muscle activations, a standard criterion for resolving muscle redundancy in musculoskeletal simulation.

The simulation outputs continuous activation trajectories for 40 bilateral shoulder muscle strands, with 20 strands on each

TABLE I
COMPARISON WITH RELATED MUSCLE ACTIVATION DATASETS.

Dataset	Year	Body Region	Subjects	Duration	Muscles	Shoulder Muscles	Actions	Activation Type	Motion Data
Camargo [2]	2021	Lower limb	22	10 min	11	None	4	Real EMG	Kinematics / GRF
KIMHu [13]	2023	Upper limb	20	10 h	4	1	3	Real EMG	RGB-D / Skeleton
MIA [14]	2023	Whole-body	10	12.5 h	8	2	15	Real sEMG	Video / Skeleton
MuscleMap [15]	2024	Whole-body	N/A	~25 h ^a	20 ^b	1	135	Binary group labels	RGB / Skeleton
MinT [5]	2024	Lower body + trunk	227	10 h	402 ^c	None	187	Simulated activation	3D pose
BioShoulder (Ours)	2026	Shoulder	302	9.73 h	40^c	40	30	Simulated activation	3D pose

^a Coarse estimation based on 15,004 clips of 3–9 s. ^b Clip-wise binary labels for muscle groups. ^c Muscle strands; some anatomical muscles may be represented by multiple strands.

side. These strands correspond to muscles acting on the shoulder joint and do not include upper-arm muscles that primarily actuate the elbow. After simulation and quality control, motion features and activation trajectories are resampled to 20 Hz during dataset construction.

B. Motion Curation and Quality Control

Directly applying musculoskeletal simulation to large-scale motion repositories can produce unreliable muscle activations when the input motions are incompatible with the assumptions of the biomechanical model. In BioShoulder, we therefore apply a multi-level curation pipeline before constructing the final motion–activation clips.

Trial-level semantic filtering. Shoulder muscle recruitment during object interaction or physical contact depends on external forces and torques that are generally not recoverable from kinematics alone. To reduce this source of ambiguity, we use the action annotations provided by BABEL to remove AMASS trials involving substantial external interactions, such as object manipulation, contact-heavy actions, or load-bearing motions. This step retains free-space upper-body motions for which the missing external-force information is less likely to dominate the simulated shoulder activations.

Trial-level simulation feasibility filtering. After semantic filtering, each retained trial is processed by inverse kinematics and static optimization in OpenSim. Trials are discarded if the IK/SO pipeline fails to complete successfully under the kinematic, joint-range, and actuator constraints defined in the bilateral MOBL-ARMS model. This filtering removes motions that are incompatible with the musculoskeletal model or lead to unsuccessful optimization, preventing such failures from entering the final dataset.

Window-level activation screening. Finally, after temporal segmentation, we screen the generated clips for implausible activation saturation. Since the retained motions are free-space movements, clips containing saturated muscle activations at the upper bound are removed from the benchmark dataset. This step is performed at the window level rather than the trial level, preserving otherwise valid trials while excluding localized abnormal segments.

These steps reduce errors caused by unmodeled external interactions, model-incompatible kinematics, and localized activation artifacts, yielding quality-controlled simulated activations for free-space shoulder motions.

TABLE II
STATISTICS OF THE FINAL AMASS SUBSETS USED IN BIO SHOULDER.

Subset	Subjects	Experiments	Windows	Duration (h)
BMLmovi	86	457	2,009	0.4795
BMLrub	87	145	8,429	1.6683
CMU	71	298	9,674	1.9404
EyesJapanDataset	12	396	15,692	3.1304
KIT	41	571	10,454	2.1449
TotalCapture	5	19	1,879	0.3692
Total	302	1,886	48,137	9.7327

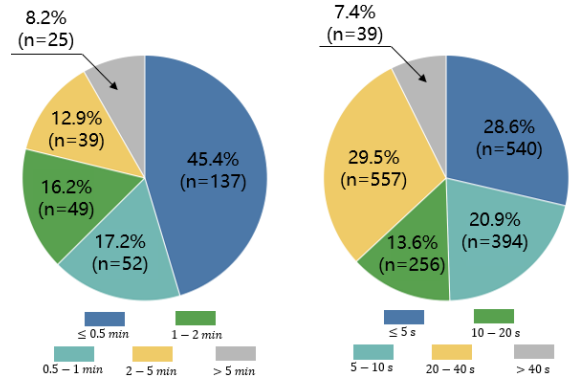


Fig. 2. Duration distributions in BioShoulder. Left: distribution of total retained motion duration per subject. Right: distribution of continuous trial durations. Numbers in parentheses indicate the number of subjects or trials in each bin, respectively.

C. Motion–Activation Pair Construction

After simulation and quality control, each continuous trial is converted into frame-level motion–activation pairs. The simulated activation trajectories and motion representations are resampled at 20 Hz and aligned temporally at the frame level. We then segment each continuous trial into 1.4-second clips with 50% overlap, yielding sequences of 28 frames.

Each clip contains the corresponding motion sequence, a 40-dimensional bilateral shoulder activation sequence, subject and trial identifiers, and retained BABEL action labels. The activation sequence is represented as a $T \times 40$ matrix, where $T = 28$ and the 40 channels correspond to 20 shoulder muscle strands on each side. The action labels are retained as metadata to support action-level querying, stratified analysis, and the construction of evaluation splits.

D. Dataset Statistics and Characteristics

The final BioShoulder dataset is constructed from six AMASS sub-datasets: EyesJapan, CMU, BMLrub, KIT, BML-movi, and TotalCapture. As summarized in Table II, BioShoulder contains 1,886 continuous free-space motion trials from 302 subjects, with a total duration of 9.732 hours. After window segmentation, these trials yield 48,137 fixed-length motion-activation clips. We distinguish continuous trials from clips throughout the paper: a trial denotes an original continuous motion sequence after filtering, whereas a clip denotes a 1.4-second overlapping segment used for sequence modeling.

At the dataset level, BioShoulder covers 30 action categories and provides continuous activations for 40 bilateral shoulder muscle strands. All motion features and activation trajectories are uniformly sampled at 20 Hz and segmented into 1.4-second clips with a stride of 0.7 s. The median trial duration is 10.20 s, while the longest retained trial lasts 214.00 s, indicating that BioShoulder contains both short isolated gestures and longer continuous upper-body movements. Figure 2 shows that most retained trials are short, indicating that BioShoulder is mainly composed of short-duration free-space motion segments.

BioShoulder covers 30 action categories after semantic curation. The largest categories include static poses, arm movements, hand movements, body-part touching, gestures, and shoulder movements, indicating that the dataset contains diverse no-contact upper-body motions. Because BABEL annotations can assign multiple labels to the same trial, category-wise durations are not mutually exclusive. Detailed category-level statistics are provided in Appendix A.

IV. EXPERIMENTS AND BENCHMARKS

To demonstrate the utility of the BioShoulder dataset, we establish a comprehensive benchmark for the task of mapping high-dimensional human motion features to continuous shoulder muscle activations. This task is crucial for developing biomechanically-aware neural networks capable of understanding internal human physical exertion.

A. Task Formulation and Experimental Setup

We formulate motion-to-muscle activation prediction as a sequence-to-sequence regression task. For each 1.4-second clip, the input is a 263-dimensional HumanML3D-style motion representation [18], denoted as $\mathbf{X} \in \mathbb{R}^{T \times 263}$, and the target is the corresponding bilateral shoulder activation sequence $\mathbf{Y} \in \mathbb{R}^{T \times 40}$. With a sampling rate of 20 Hz, each clip contains $T = 28$ frames. The benchmark includes 48,137 clips derived from 1,886 continuous source sequences.

We evaluate all models under two complementary protocols. In the *sequence-level split*, source sequences are partitioned into training, validation, and test sets, containing 37,597/5,354/5,186 clips from 1,508/189/189 source sequences, respectively. This protocol evaluates prediction on held-out motion sequences drawn from the overall action distribution. In the more challenging *motion-type split*, the sets contain mutually exclusive primary action types. The training, validation, and test sets comprise 38,509/4,814/4,814

clips from 1,279/393/214 source sequences, respectively. This protocol evaluates generalization to action types not observed during training.

We compare four representative sequence-modeling architectures: LSTM [19], FConv [20], Mamba2 [21], and Transformer [22]. All models are trained from scratch using the Smooth L1 loss and AdamW with a learning rate of 10^{-3} , weight decay of 10^{-4} , and batch size 128. Training runs for at most 500 epochs with a 5% linear warm-up followed by cosine decay and early stopping with a patience of 10 epochs. We conduct five runs using seeds {1, 42, 133, 1024, 2026} and report the mean and sample standard deviation.

B. Evaluation Metrics

Following MinT [5], we evaluate muscle activation prediction using Root Mean Squared Error (RMSE), Symmetric Mean Absolute Percentage Error (SMAPE), and Pearson Correlation Coefficient (PCC). RMSE measures absolute prediction error but is sensitive to signal scaling. PCC evaluates activation-series similarity while being invariant to scale and offset, whereas SMAPE retains sensitivity to fixed offsets while reducing the influence of overall scaling. PCC and SMAPE are computed independently for each of the 40 shoulder muscle strands and then averaged. We report RMSE in units of 10^{-3} , SMAPE as a percentage, and PCC on its original scale.

C. Results and Discussion

Table III reports the results under both evaluation protocols. The Transformer achieves the strongest overall performance on the sequence-level split, obtaining an RMSE of 19.891×10^{-3} , a SMAPE of 18.914%, and a PCC of 0.697. Mamba2 provides the second-best overall results. FConv outperforms LSTM in RMSE and PCC but produces a higher SMAPE, illustrating that absolute error, relative error, and temporal similarity capture complementary aspects of activation prediction. All models perform worse under the motion-type split, confirming that generalization to unseen primary motion types is more challenging than prediction on held-out source sequences. Nevertheless, the Transformer remains the best-performing model across all three metrics, achieving an RMSE of 24.123×10^{-3} , a SMAPE of 21.114%, and a PCC of 0.640. These results suggest that attention-based temporal modeling is effective for learning the motion-dependent activation patterns represented in BioShoulder.

Figure 3 and Figure 4 present two representative examples from the sequence-level test set. In both figures, solid black curves denote simulated ground-truth activations, and dashed blue curves denote Transformer predictions. In the clapping motion, the selected shoulder muscle activations exhibit clear bilateral and periodic patterns, especially in the DELT1 strands on both sides, which is consistent with the repeated arm movement involved in clapping. In contrast, the pointing motion shows a more asymmetric activation pattern: the left shoulder remains largely inactive, whereas the right shoulder produces isolated, non-periodic activation peaks across

TABLE III

BENCHMARK RESULTS FOR MAPPING HIGH-DIMENSIONAL MOTION FEATURES TO SHOULDER MUSCLE ACTIVATIONS UNDER SEQUENCE-LEVEL AND MOTION-TYPE SPLITS.

Dataset split	Model	RMSE (10^{-3}) ↓	SMAPE (%) ↓	PCC ↑
Sequence-level	LSTM [19]	22.201 ± 0.132	21.411 ± 0.366	0.586 ± 0.005
	FConv [20]	20.472 ± 0.140	22.613 ± 0.217	0.674 ± 0.004
	Mamba2 [21]	20.229 ± 0.228	20.156 ± 0.401	0.680 ± 0.009
	Transformer [22]	19.891 ± 0.157	18.914 ± 0.572	0.697 ± 0.009
Motion-type	LSTM [19]	26.338 ± 0.166	23.792 ± 0.159	0.539 ± 0.006
	FConv [20]	25.064 ± 0.159	24.596 ± 0.177	0.606 ± 0.010
	Mamba2 [21]	24.579 ± 0.131	22.876 ± 0.702	0.621 ± 0.002
	Transformer [22]	24.123 ± 0.222	21.114 ± 0.358	0.640 ± 0.011

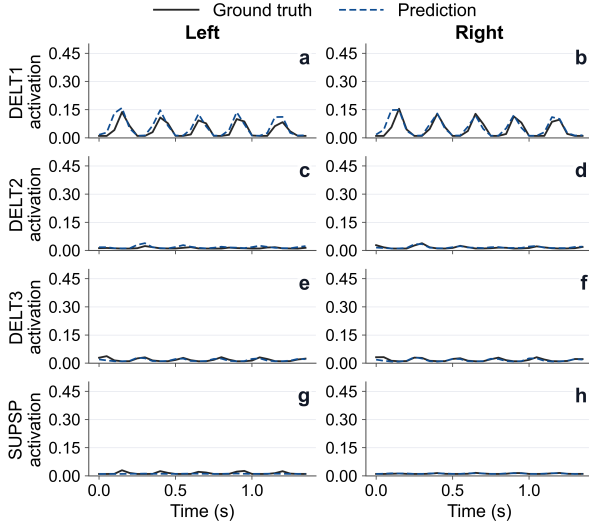


Fig. 3. Transformer prediction example for a clapping motion, showing bilateral and periodic shoulder activation patterns.

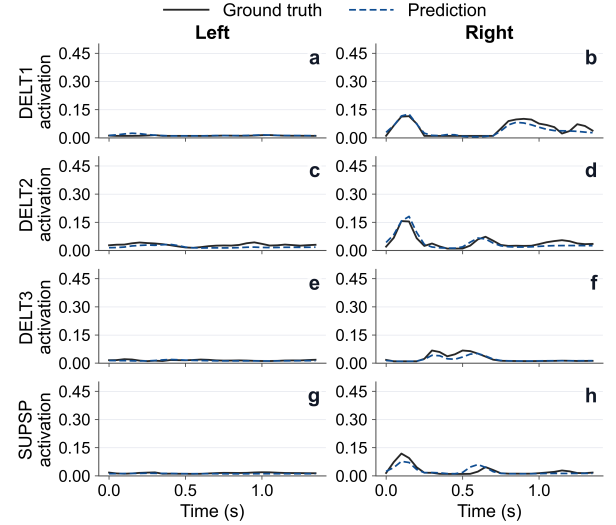


Fig. 4. Transformer prediction example for a pointing motion, showing asymmetric and right-dominant shoulder activation patterns.

several selected strands. The Transformer predictions capture the main temporal structure, laterality, and active/inactive muscle patterns in both examples, although some amplitude deviations remain around peak activations. These observations provide qualitative evidence that the simulated activations contain motion-dependent and biomechanically interpretable recruitment patterns that can be learned from the paired motion data.

V. CONCLUSION AND LIMITATIONS

We introduced BioShoulder, a large-scale dataset pairing human motion with continuous bilateral shoulder muscle activations. The dataset contains 1,886 continuous trials and 48,137 fixed-length clips with 40 simulated shoulder activation channels. Trial-level semantic curation, simulation feasibility filtering, and window-level activation screening reduce common sources of simulation error. Our benchmarks demonstrate the utility of BioShoulder for motion-to-muscle prediction: the Transformer achieves the strongest overall performance, while the consistent performance decline under the motion-type split highlights the difficulty of generalizing to unseen

motion types. The dataset, official splits, and benchmark code are publicly available.

Despite its quality-control pipeline, BioShoulder has two main limitations. First, because the activations are derived from musculoskeletal simulation, a simulation-to-real gap may remain when applying models trained on BioShoulder to real physiological settings. The generated activations also depend on the anatomical parameterization of MOBL-ARMS and the muscle-recruitment assumptions of static optimization. Second, BioShoulder focuses on free-space motions because the external forces and torques involved in object interaction or load-bearing actions cannot be reliably inferred from kinematics alone. Consequently, the dataset does not currently support applications that require explicit modeling of human-object interaction or external loads.

APPENDIX A ACTION CATEGORY STATISTICS

Table IV reports the duration and number of continuous trials associated with each retained action category. Since a trial may contain multiple BABEL labels, category-wise durations and counts are not mutually exclusive.

TABLE IV
CATEGORY-LEVEL STATISTICS OF RETAINED ACTION LABELS.

Category	Duration (min)	Trials	Category	Duration (min)	Trials
t-pose	369.98	697	flap	9.56	17
arm movements	284.25	755	grab body part	8.22	17
hand movements	237.35	654	point	6.74	36
touching body part	129.00	357	salute	4.20	13
gesture	80.56	260	style hair	2.55	5
shoulder movements	65.77	52	clasp hands	1.88	4
wrist movements	64.01	51	juggle	1.81	4
upper body movements	63.30	59	draw	1.35	4
wave	53.39	210	shrug	1.28	6
touching face	44.61	84	sign	1.17	6
greet	37.18	187	flail arms	1.14	3
clap	29.64	104	groom	0.74	1
scratch	25.75	104	stroke	0.62	2
crossing limbs	21.02	83	slash gesture series	0.41	1
a pose	18.67	90	count	0.09	1

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